F 03-09

NON-CONFORMANCE, CORRECTIVE ACTION AND CLEARANCE REPORT

SANAS No.:

Organisation Name: Azwedy PTY LTD

NC Report Number: NC014-2026

Area/Scope Assessed: System documentation

Assessor / Inspector / Expert: Mapanya Tshepo

PART 1: DETAILED OBSERVATION

There is no clear indication of contributors to measurement uncertainty. UB001 only indicates Expanded uncertainty, no contributors mentioned.

Signed (Representative):

Signed (Team Leader / Inspector):

MAJOR: ☑ MINOR: ☐

Reference No. of relevant Guide / Standard / Requirement:

Date: 12/02/2026

PART 2: ROOT CAUSE ANALYSIS

(Provide an analysis of the extent and cause of the nonconformity)

Oversight, incomplete characterization of the validation (uncertainty budget) process where method-specific contributors were omitted from the UB001 uncertainty budget

PART 3: CORRECTIVE ACTION TAKEN

(Clearly indicate what corrective action was taken, and attach supporting evidence of implementation)

Revised UB001 to include detailed uncertainty budget tables for Volatile matter, Inherent moisture, Ash determination and calorific value. Each table now explicitly lists both type A and type B contributors with their respective data sources.

Date: 12/02/2026

Signed (Representative):

PART 4: CORRECTIVE ACTION VERIFIED AND CLEARED

(Comment on whether the corrective action is sufficient and appropriate)

Addressed

Date: 12/02/26

Signed: Assessor / Inspector/Expert:

PART 5: FOLLOW-UP AT NEXT ASSESSMENT

(Comment on the implementation of the corrective actions)

Date:

Signed: Assessor / Inspector/Expert:

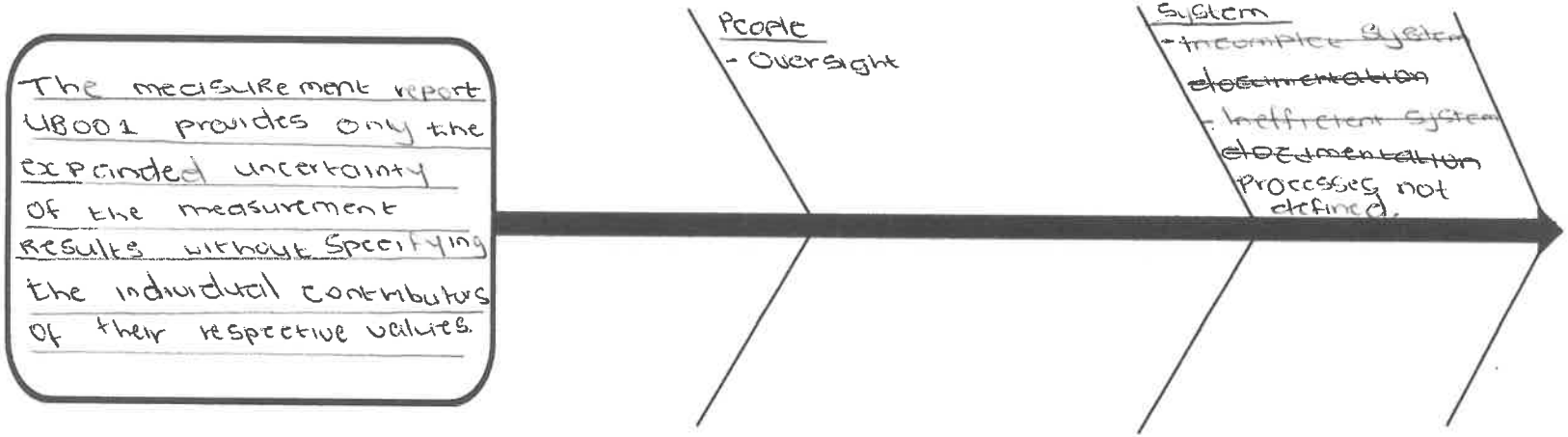
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AS001MF

07/25

ROOT CAUSE ANALYSIS FORM

IRQ NO: NC014-2026



SUPERVISOR

AS004MF 07/25

BRAINSTORMING FORM

IRQ no: NC014-2026

Brainstorming team
Nosipho Hlongwane, Keitumetse Maleka, Abenathi Jama and Sifiso Nyambi

What went wrong?

The measurement report UB001 provides only the expanded uncertainty of the measurement results without specifying the individual contributors or their respective values.

Why 1: Why does UB001 only report expanded uncertainty without documenting the individual uncertainty contributors?

- Incomplete characterisation of the validation (Uncertainty budget) process.

Why 2: Why was there an incomplete characterisation of the validation process?

- Oversight, method - specific contributors were omitted from UB001 uncertainty budget.

Supervisor:

Before

AzurESky "ensure quality of test"	AzureSky		
	Uncertainty Budget		Document no: UB001
	Compiled by:	Sifiso Nyambi - Laboratory Manager	
	Approved by:	Nonhlanhla Msibi - Operations Manager	

AZURESKY (PTY) LTD

MASTER MEASUREMENT UNCERTAINTY BUDGET

(ISO/IEC 17025:2017 Clause 7.6 & SANAS F49 Requirements)

1. Purpose and Scope

This consolidated uncertainty budget encompasses **all calibrated instruments** used in the analytical determination of moisture, ash, volatile matter, fixed carbon, and calorific value.
All calibrations are traceable to **NMISA** through **SANAS-accredited laboratories** (Micron Lab & Montech Calibration Services).

2. The Crux / Nucleus of Method Validation

- **Performance characteristics** (range, accuracy, precision, repeatability) are clearly established.
- **Methods and instruments are validated and verified** for representative coal sample types.
- **Results meet customer and method specifications,** ensuring traceable, reliable measurements.

Effective date	Rev status	Next review date
07/2025	00	07/2028

Before

AzurESky ensure quality of test	AzureSky		
	Uncertainty Budget	Document no: UB001	
	Compiled by:	**Sifiso Nyambi - Laboratory Manager**	
	Approved by:	**Nonhlanhla Msibi - Operations Manager**	

3. General Calibration Information

Customer	**AzureSky (Pty) Ltd – Unit 3, 17 Markiel Rd, Wadeville, Germiston 1401**
Contact Person	nonhlanhla@azuresky.it.com
Calibration Providers	Micron Lab (Pty) Ltd – SANAS Accredited. No. 2017/365305/07 Montech Calibration Services (Pty) Ltd – SANAS Accredited
Traceability	All results traceable to NMISA (South Africa)
Confidence Level	Expanded uncertainty with coverage factor *k = 2 (≈ 95 % confidence)*
Decision Rule	Within Tolerance / Out of Tolerance / Indeterminate (as per SANAS F49)

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07/2025	00	07/2028

Before

AzureSky		
Uncertainty Budget	**Document no: UB001**	
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Approved by:	**Nonhlanhla Msibi - Operations Manager**	

4. Consolidated List of Calibrated Instruments

Instrument	Manufacturer / Model	Serial No.	Certificate No.	Date of Calibration	Next Due	Expanded Uncertainty (k = 2)	Status
Analytical Balance	BEL M124Ai	ITA2500477	SET 3	25 Mar 2025	Mar 2026	± 0.0004 g	Within Tolerance
Top Pan Balance	AE Adam CKT8H	AEA90242156	AZU T1.1 (CAL 103-03-00 / 14-00)	Mar 2025	Mar 2026	± 0.3 g	Within Tolerance
Platform Scale	Darvan Brenco 150 kg	120330150	SET 5 (25.03.2025)	25 Mar 2025	Mar 2026	± 0.03 kg	Within Tolerance
Furnace (Volatile)	ATS 210325/59V	–	AZU VF1.1 (CAL 103)	Mar 2025	Mar 2026	± 4 °C	Within Tolerance
Furnace (Ashing)	ATS 210325/74A	–	CAL 103	Mar 2025	Mar 2026	± 4 °C	Within Tolerance

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Instrument	Manufacturer / Model	Serial No.	Certificate No.	Date of Calibration	Next Due	Expanded Uncertainty (k = 2)	Status
Oven	Darvan Brenco 50 L	DOV53A/210325/40	CAL 103	Mar 2025	Mar 2026	± 2 °C	Within Tolerance
Thermo-Hygrometer	Sinometer SDC103	OBS 202503-01	HI253D13-31 (Montech)	17–18 Mar 2025	18 Mar 2026	± 0.95 % RH / ± 0.2 °C	Within Tolerance
Digital Thermometer + Probe	Sinometer DM6801B	S1208105	OBS T1.1a (Revised)	14 Mar 2025	Mar 2026	± 0.3 °C	Within Tolerance
Weight Set 1	OB WS1 (9 weights, 1 g – 100 g)	–	OB WS1.1R	17 Mar 2025	31 Mar 2026	± 0.00005 – 0.0001 g	Within Tolerance
Weight Set 2	OB WS2 (4 weights, 5 kg – 20 kg)	–	OB WS2.1R	17 Mar 2025	31 Mar 2026	± 0.09 – 0.5 g	Within Tolerance

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Weight Set 2	OB WS2 (4 weights, 5 kg – 20 kg)	–	OB WS2.1R	17 Mar 2025	31 Mar 2026	± 0.09 – 0.5 g	Within Tolerance

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5. Uncertainty Budget Summaries

5.1 Balances and Weights

Balances and weights are evaluated jointly since mass calibration forms a single chain of traceability.

Source	Estimate	Distribution	Divisor	u(xi)	Type	Notes
Calibration (Weights + Balance)	0.0004 g – 0.3 g	Normal	2	0.0002 – 0.15	B	From SANAS certificates OB WS1.1R & WS2.1R
Repeatability	0.0001 g – 0.1 g	Normal	1	0.0001 – 0.1	A	10 replicates
Resolution	Half readability	Rectangular	√3	0.00006 – 0.06	B	Instrument display
Air Buoyancy Effect	Negligible	Rectangular	√3	–	B	For small weights only

Expanded U (k = 2):

- Analytical Balance = ± 0.0005 g
- Top Pan Balance = ± 0.4 g
- Platform Scale = ± 0.04 kg

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- Weights = ± 0.0001 g to ± 0.5 g

5.2 Temperature-Controlled Equipment

(Ovens and Furnaces, including Digital Thermometer Probe)

Source	Estimate	Distribution	Divisor	u(xi)	Type	Notes
Controller Resolution	1 °C	Rectangular	√3	0.58	B	Display limit
Sensor Drift	1 °C	Rectangular	√3	0.58	B	Thermocouple stability
Mapping Variation	1 °C	Normal	1	1	A	Temp. mapping
Calibration Certificate	2–4 °C	Normal	2	1–2	B	SANAS certificates CAL 103 / OBS T1.1a

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Expanded U (k = 2):

- Oven = ± 3 °C Furnaces = ± 5 °C Thermometer = ± 0.3 °C

5.3 Thermo-Hygrometer

Parameter	Source	Estimate	Distribution	u(xi)	Type	Notes
Temperature (°C)	Certificate HI253D13-31	± 0.2 °C	Normal	0.1 °C	B	At 22.5 °C, 47 % RH
Humidity (% RH)	Certificate HI253D13-31	± 0.95 % RH	Normal	0.48 % RH	B	Range 15 – 77 % RH
Repeatability	0.1 °C / 0.2 % RH	Normal	1	0.1 °C / 0.2 % RH	A	From replicates under lab conditions

Expanded U (k = 2): ± 0.2 °C and ± 0.95 % RH

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6. Combined Analytical Uncertainty

Propagation of the individual equipment uncertainties shows that total analytical uncertainty remains well below the precision limits of ISO methods 589, 1171, 562, 5925 and ISO 1928.
Therefore, measurement uncertainty does not materially affect analytical results, confirming fitness for purpose.

7. Traceability and Compliance

All instruments are traceable to national standards through SANAS-accredited calibration.
Each certificate explicitly references **ISO/IEC 17025** compliance, ensuring an **unbroken chain of traceability** to the **SI units**.

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8. Conclusion

The integrated uncertainty budget demonstrates that:

- All instruments are within tolerance, fit for use, and traceable.
- The combined expanded uncertainty at 95 % confidence satisfies ISO/IEC 17025 Clause 7.6.
- The laboratory maintains a single, structured **uncertainty framework** suitable for continuous expansion as new equipment is introduced.

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AzureSky		
Uncertainty Budget	**Document no: UB001**	
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9. Calorific Value Specific Uncertainty budget

Contributor	Estimate / Value	Distribution	Divisor	Standard Uncertainty u(xi)	Type	Data Source
Balance Calibration	0.0004g	Normal	2	0.0002g	B	**UB001 Sec 4** (Analytical Balance)
Digital Thermometer	0.3°C	Normal	2	0.15°C	B	**UB001 Sec 4** (Probe Calibration)
Method Repeatability	0.0144MJ/kg	Normal	1	0.0144MJ/kg	**A**	**AS002MV Sec 3.4.1** (Validation Data)
CRM Homogeneity	0.0117MJ/kg	Normal	1	0.0117 MJ/kg	**A**	**AS002MV Sec 3.4.1** (In-house CRM)
CRM Calibration/Value	0.1083 MJ/kg	Normal	2	0.0542MJ/kg	**B**	**AS002MV Sec 3.4.1** (In-house CRM)
Bomb Stability	Negligible	Rectangular		-	B	**AS002MV Sec 3.4.1**

Effective date	Rev status	Next review date
02/2026	01	02/2029